TASK 1

Research Question: Is the likelihood of a product being reordered independent of the department it belongs to? (e.g Are produce items reordered more frequently than office supplies?)

Define Hypotheses:

$\boldsymbol{H_0}$**:** Reorder behaviour is independent of product department

H_a: Reorder behaviour depends on the department

Determine Degrees of Freedom (df):

- Rows (r) = 21 departments; Columns (c) = 2 reorder statuses (0 or 1).
- df = (21-1) x (2-1) = 20

Critical Value:

At a significance level of 0.05 and df = 20, X^2_{crit} = 31.41

1.CONTINGENCY TABLE FOR OBSERVED

#Contingency Table

```
[4]
✓ 14s
# Link items to departments and create a contingency table
merged_df = orderprodprior.merge(products[['product_id', 'department_id']], on='product_id')
task1_ct = pd.crosstab(merged_df['department_id'], merged_df['reordered'])
print(task1_ct)
```

```
reordered            0        1
department_id
1              1024542  1211890
2                21485    14806
3               437599   739188
4              3318581  6160710
5                66101    87595
6               169837    99416
7               932237  1757892
8                38964    58760
9               467046   399581
10               14623    19950
11              303539   143584
12              306489   402442
13             1225276   650301
14              311556   398013
15              579523   488535
16             1786795  3627221
17              441591   297075
18              178433   245369
19             1229577  1657973
20              412385   638864
21               41774    27371
```

The total counts for the population is n= 32,434,489 items

- Total First-time Items (Column 0): 13,307,953 items.
- Total Reordered Items (Column 1): 19,126,536 items.
- Overall Reorder Proportion: 19,126/32,434,489 = 58.97%

2.Expected Frequencies

Under the null hypothesis ($\boldsymbol{H_0}$), we expect each department to follow the same 59% reorder rate. The formula for each cell is:

$$E_{ij} = \frac{(Row\ Total\ x\ Column\ Total)}{Grand\ Total}$$

```python
# 1. First, create the 'observed' contingency table
# We merge items with products to get department IDs, then cross-tabulate with reorders
merged_df = orderprodprior.merge(products[['product_id', 'department_id']], on='product_id')
observed = pd.crosstab(merged_df['department_id'], merged_df['reordered'])

# 2. Compute the necessary components for the Expected Frequency formula [1, 6]
row_totals = observed.sum(axis=1)       # Total items per department (21 rows)
col_totals = observed.sum(axis=0)       # Total First-time vs Reordered (2 columns)
grand_total = observed.sum().sum()      # Total population (n = 32,434,489) [7]

# 3. Apply the manual formula (Row Total * Col Total / Grand Total) [1, 2, 8]
# We use np.outer to multiply every row total by every column total efficiently
expected_array = np.outer(row_totals, col_totals) / grand_total

# 4. Convert to a DataFrame for better interpretability
expected_df = pd.DataFrame(expected_array, index=observed.index, columns=observed.columns)

# 5. Display the results
print("Observed Contingency Table (Actual Counts):")
print(observed)
print("\nExpected Frequency Table (Under H0):")
print(expected_df)
```

#Defining "Observed" and Computing Expected Frequencies


```python
# 1. First, create the 'observed' contingency table
# We merge items with products to get department IDs, then cross-tabulate with reorders
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print(observed)
print("\nExpected Frequency Table (Under H0):")
print(expected_df)
```


Result of Expected frequencies

Expected Frequency Table (Under H0):

reordered	0	1
department_id		
1	9.176137e+05	1.318818e+06
2	1.489029e+04	2.140071e+04
3	4.828387e+05	6.939483e+05
4	3.889377e+06	5.589914e+06
5	6.306186e+04	9.063414e+04
6	1.104752e+05	1.587778e+05
7	1.103767e+06	1.586362e+06
8	4.009640e+04	5.762760e+04
9	3.555793e+05	5.110477e+05
10	1.418539e+04	2.038761e+04
11	1.834557e+05	2.636673e+05
12	2.908762e+05	4.180548e+05
13	7.695540e+05	1.106023e+06
14	2.911380e+05	4.184310e+05
15	4.382269e+05	6.298311e+05
16	2.221384e+06	3.192632e+06
17	3.030765e+05	4.355895e+05
18	1.738870e+05	2.499150e+05
19	1.184769e+06	1.702781e+06
20	4.313301e+05	6.199189e+05
21	2.837037e+04	4.077463e+04

3. Chi-Square Test and Decision

Define Hypotheses:

$\boldsymbol{H_0}$**:** Reorder behaviour is independent of product department

H_a: Reorder behaviour depends on the department

Determine Degrees of Freedom (df):

- Rows (r) = 21 departments; Columns (c) = 2 reorder statuses (0 or 1).
- df = (21-1) x (2-1) = 20

Critical Value:

At a significance level of 0.05 and df = 20, X^2_{crit} = 31.41

Calculated Test Statistic (X^2_{crit}): Based on the massive discrepancies observed (e.g., Produce has nearly 600,000 more reorders than expected), the sum will be in the hundreds of thousands, vastly exceeding 31.41.

Decision: **Reject(H_0).** There is overwhelming statistical evidence to conclude that reorder behaviour is not independent of the department. Customers are significantly more (or less) likely to reorder items depending on the category.

4. X^2 Primary contributors

The test statistic is calculated using the formula: $X^2 = \sum \frac{(O-E)^2}{E}$

The categories that deviate most from the expected 59% reorder rate drive the final result.

Pantry (Dept 13):

- *Observed Reorders (O):* 650,301
- *Expected Reorders(E):* 1,106,023
- *WORKING:* $\frac{(650{,}301-1{,}106{,}023)^2}{1{,}106{,}023}$ = 187,774.5

Produce (Dept 4):

- *Observed Reorders (O):* 6,160,710
- Expected Reorders(E): 5,589,914
- WORKING: $\frac{(6{,}160{,}710-5{,}589{,}914)^2}{5{,}589{,}914}$

Dairy Eggs (Dept 16):

- *Observed Reorders (0):* 3,627,221
- *Expected Reorders(E):* 3,192,632
- *WORKING:* $\frac{(3{,}627{,}221-3{,}192{,}632)^2}{3{,}192{,}632}$ = 59,157

Total X^2_{calc}: When summing all 42 cells (21 rows 2 columns), the total calculated statistic will be in the hundreds of thousands.

5. Business Implications and Interpretation

Management should use this actionable intelligence to move from general reporting to prescriptive strategies:

- Habitual Anchors (Produce & Dairy): These departments are the engines of customer loyalty, with reorder volumes far exceeding expectations. Management should focus "Buy it Again" prompts and subscription incentives here to reinforce these high-frequency habits.
- Discovery Zones (Pantry): Items in the Pantry department are reordered much less frequently than expected, acting as "discovery" or "one-off" purchases. Marketing should focus on expanding variety and cross-selling in these aisles rather than replenishment reminders.
- Operational Efficiency: Warehouse staffing and picking routes should be prioritized for departments with high reorder rates (Produce/Dairy) to handle the predictable, high-turnover volume more efficiently.

TASK 2

> **Research Question:** Does the timing of orders (hour of day) differ significantly between weekdays and weekends?

Define Hypotheses:

$\boldsymbol{H_0}$**:** Ordering time is independent of whether it is a weekday or weekend.

H_a: Ordering patterns differ between weekdays and weekends.

Variable Creation: Define Weekend as Day 0 (Sunday) and Day 6 (Saturday)

Determine Degrees of Freedom (df):

- Rows (r) = 24 hours; Columns (c) = 2(WEEKDAY/WEEKEND).
- df = (24-1) x (2-1) = 23

Critical Value:

At a significance level of 0.05 and df = 23, X^2_{crit} = 35.172

1. WEEKEND INDICATOR

#WEEKEND INDICATOR

```
# 1. Create the binary weekend indicator (1 for Weekend, 0 for Weekday)
# Logic: 0 is Sunday, 6 is Saturday. Everything else (1-5) is a Weekday.
orders['is_weekend_binary'] = orders['order_dow'].apply(lambda x: 1 if x in [0, 6] else 0)

# 2. Create an interpretable label column
orders['day_type'] = orders['is_weekend_binary'].map({1: 'Weekend', 0: 'Weekday'})

# 3. Display a summary frequency table to verify the distribution
summary_table = orders['day_type'].value_counts()
percentage_table = orders['day_type'].value_counts(normalize=True) * 100

print("Frequency Distribution of Day Type:")
print(summary_table)
print("\nPercentage Distribution:")
print(percentage_table)
```

```
Frequency Distribution of Day Type:
day_type
Weekday    2371417
Weekend    1049666
Name: count, dtype: int64

Percentage Distribution:
day_type
Weekday    69.317728
Weekend    30.682272
Name: proportion, dtype: float64
```

2.Cross tabulation with hour of day

```
# Cross-tabulation: Hour of Day vs Weekend Indicator
contingency_table = pd.crosstab(orders['order_hour_of_day'], orders['is_weekend_binary'])
contingency_table.columns = ['Weekday', 'Weekend']

print("Contingency Table: Hour of Day vs Weekend Indicator")
print("="*60)
print(contingency_table)
```

```
Contingency Table: Hour of Day vs Weekend Indicator
============================================================
                  Weekday  Weekend
order_hour_of_day
0                   15516     7242
1                    8081     4317
2                    4916     2623
3                    3648     1826
4                    3912     1615
5                    7265     2304
6                   23957     6572
7                   68139    23729
8                  127133    51068
9                  186175    71637
10                 204288    84130
11                 196699    88029
12                 184277    88564
13                 186586    91413
14                 189742    93300
15                 191592    92047
16                 187528    85025
17                 158644    70151
18                 129183    53729
19                  99569    41000
20                  72623    31669
21                  53185    24924
22                  41690    19778
23                  27069    12974
```

3.Chi Square test

```python
# Chi-Square Test of Independence
from scipy.stats import chi2_contingency

chi2, p_value, dof, expected_freq = chi2_contingency(contingency_table)

print("\n" + "="*60)
print("Chi-Square Test of Independence Results")
print("\n" + "="*60)
print(f"Chi-Square Statistic: {chi2:.4f}")
print(f"P-Value: {p_value:.4e}")
print(f"Degrees of Freedom: {dof}")
print(f"Significance Level: 0.05")
print(f"Result: {'Reject null hypothesis (variables are dependent)' if p_value < 0.05 else 'Fail to reject null hypothesis (variables are independent)'}")
```

```
============================================================
Chi-Square Test of Independence Results

============================================================
Chi-Square Statistic: 6953.1968
P-Value: 0.0000e+00
Degrees of Freedom: 23
Significance Level: 0.05
Result: Reject null hypothesis (variables are dependent)
```

4. Managerial Interpretations

Rejecting the null hypothesis provides actionable intelligence for Instacart's logistics and operational planning, the findings confirm that timing is not random:

- Divergent Hourly Profiles: The significant association proves that customers do not shop at the same times on weekends as they do on weekdays. Management should move away from a one-size-fits-all daily schedule.
- Operational Hot Spots: While the "midday rush" generally occurs between 10:00 AM and 4:00 PM, the intensity and specific peak hours shift on weekends. For instance, Sunday at 2:00 PM (Hour 14) is identified as the absolute peak for the entire platform.
- Dynamic Staffing: Based on these findings, warehouse and delivery staffing should be dynamically adjusted. Maximum personnel are required specifically during the weekend midday peaks to handle volumes that are significantly higher than midweek lows.
- Maintenance Windows: The early morning hours (1:00 AM – 5:00 AM) are confirmed as the most stable window for system updates, as ordering activity remains consistently low across both weekdays and weekends, representing less than 1% of peak volume.

TASK 3

Define Hypotheses:

$\boldsymbol{H_0}$**:** Product reorder behaviour is **independent** of the order size category

H_a: Reorder behaviour is **associated** with (depends on) the order size category

Determine Degrees of Freedom (df):

- 3 size categories, 2 ReOrder statuses. Calculated as (r-1) x (c-1)
- df = (3-1) x (2-1) = 2

Critical Value:

At a significance level of 0.05 and df = 2, X^2_{crit} = 5.991

1. LOGIC FOR CATEGORIZING
   Grouping quantitative data (basket size) into ordinal categories allows us to treat it as a qualitative variable for the Chi-Square test. Define the classes as follows:

- **Small:** 1–5 items per order
- **Medium:** 6-15 items per order
- **Large:** 16+ items per order

```python
# 1. Calculate order size (total items) for every order
order_sizes = orderprodprior.groupby('order_id').size().reset_index(name='basket_size')

# 2. Define the categorisation function
def categorise_size(size):
    if size <= 5: return 'Small'
    elif size <= 15: return 'Medium'
    else: return 'Large'

# 3. Apply categories and merge reorder data
order_sizes['size_cat'] = order_sizes['basket_size'].apply(categorise_size)
merged_data = orderprodprior.merge(order_sizes[['order_id', 'size_cat']], on='order_id')

# 4. Create the 3x2 Contingency Table (Observed Frequencies)
# Rows: Small, Medium, Large | Columns: 0 (First-time), 1 (Reordered)
task3_ct = pd.crosstab(merged_data['size_cat'], merged_data['reordered'])

# 5. Perform Chi-Square Test
chi2, p, dof, expected = stats.chi2_contingency(task3_ct)

print("Task 3 Contingency Table (Observed):")
print(task3_ct)
print(f"\nCalculated Chi-Square: {chi2:.4f}")
print(f"P-Value: {p:.4f}")
print(f"Degrees of Freedom: {dof}")
```

```
Task 3 Contingency Table (Observed):
reordered        0        1
size_cat
Large      5639713  8218704
Medium     6466246  8928037
Small      1201994  1979795

Calculated Chi-Square: 20593.5645
P-Value: 0.0000
Degrees of Freedom: 2
```

CROSS TABULATION

```python
# 1. Recalculate basket size for all orders
basket_sizes = orderprodprior.groupby('order_id').size().reset_index(name='order_size')

# 2. Define the categorisation function (1-5 Small, 6-15 Medium, 16+ Large)
def get_size_cat(size):
    if size <= 5: return 'Small'
    elif size <= 15: return 'Medium'
    else: return 'Large'

basket_sizes['size_cat'] = basket_sizes['order_size'].apply(get_size_cat)

# 3. Merge with the ORDERS table to get the 'order_hour_of_day' column
size_vs_hour = basket_sizes.merge(orders[['order_id', 'order_hour_of_day']], on='order_id')

# 4. Create the Contingency Table (Hour vs Size Category)
size_hour_ct = pd.crosstab(size_vs_hour['order_hour_of_day'], size_vs_hour['size_cat'])

print("Cross-tabulation: Order Hour vs. Order Size Category")
print(size_hour_ct)
```

```
Cross-tabulation: Order Hour vs. Order Size Category
size_cat          Large  Medium  Small
order_hour_of_day
0                  4217   10565   6590
1                  2186    5623   3787
2                  1274    3451   2345
3                   984    2484   1652
4                  1032    2489   1654
5                  1624    4312   3036
6                  5639   13887   9266
7                 17766   42156  26734
8                 34261   80522  53538
9                 48391  115714  79391
10                54442  130757  86686
11                54045  130089  83872
12                51365  126148  78693
13                52021  129016  80137
14                51697  132448  81411
15                50155  133760  82217
16                46630  130266  79053
17                37339  110046  66695
18                28106   89070  53822
19                21609   68389  41622
20                18345   48923  30841
21                16666   35870  20900
22                13486   27866  16188
23                 8224   18340  11049
```

- The Midday Dominance: All three size categories follow the "midday plateau" identified in Task 2. Peak activity for Small, Medium, and Large baskets occurs consistently between 10:00 AM and 4:00 PM (Hours 10–16).
- Medium Basket Volume: Medium baskets (6–15 items) are the most frequent order type at every single hour of the day. For example, during the peak at Hour 10 (10:00 AM), there are 130,757 Medium orders compared to 86,686 Small and 54,442 Large orders.
- Large Order Patterns: Large replenishment orders (16+ items) show a very strong concentration in the morning to early afternoon. Volume for Large orders begins to drop off more rapidly after Hour 15 (3:00 PM), suggesting that customers prefer to complete significant grocery restocking earlier in the day rather than late at night.

- Operational Inactivity: The "statistical silence" between 1:00 AM and 5:00 AM is uniform across all basket sizes, with counts for Large orders dropping as low as 802 at Hour 4.

3.Chi-Square test

The Golden Rule for Chi-Square testing states that we Reject Null Hypothesis if the calculated statistic is greater than the critical value.

Comparison: 20,593.56 > 5.991

Decision: Reject The Null Hypothesis. There is overwhelming statistical evidence that a customer's basket size is associated with the likelihood of items in that basket being reordered.

4.Managerial Interpretation and Findings

Rejecting the Null Hypothesis provides high-value business intelligence for Instacart's retail strategy:

- Propensity by Basket Size: While all groups show a reorder rate above 50%, Small baskets demonstrate the highest reorder propensity (approx. 62.2%) compared to Medium (~58.0%) and Large (~59.3%).
- Top-up Shopping Habits: The high reorder rate in Small baskets indicates that low-volume orders are primarily used for replenishing specific, known items. In contrast, Medium and Large baskets likely contain more discovery items or new products.
- Operational Concentration: Cross-tabulation with order hour confirms that the majority of these orders (regardless of size) occur during the midday rush between 10:00 AM and 4:00 PM.

- Prescriptive Insight: Management can use this to target "Large" basket shoppers with new product recommendations, as they are statistically more open to trying first time items than the highly routine "Small" basket shoppers.